QRAEMON

Market Research & Competitive Intelligence

Physical AI, Robotics Intelligence,
Robot Memory and Continual Experience

Market Thesis

The emerging opportunity for a persistent experience layer
between general robot intelligence and physical deployment

Strategic Research Document

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Contents

1 Executive Summary **3**
1.1 Market Conclusion . . . 3

2 Market Structure **3**

3 Robotics Market Environment **4**
3.1 Industrial Robot Base . . . 4
3.2 Market Direction . . . 4

4 Physical AI Landscape **4**

5 Competitive Landscape **5**
5.1 Competitive Categories . . . 5

6 Key Competitor Profiles **5**
6.1 NVIDIA . . . 5
6.2 Google DeepMind . . . 6
6.3 Physical Intelligence . . . 6
6.4 Figure . . . 6
6.5 Intrinsic . . . 6

7 Competitive Positioning **7**

8 Customer Segmentation **7**

9 Geographic Strategy **7**
9.1 India . . . 7
9.2 United States . . . 8
9.3 Europe . . . 8

10 Market Size Framework **8**
10.1 Installed-Robot Model . . . 9

11 SAM and SOM Framework **9**
11.1 Serviceable Available Market . . . 9
11.2 Serviceable Obtainable Market . . . 9

12 Major Market Trends **10**
12.1 Trend 1: Physical AI . . . 10
12.2 Trend 2: General-Purpose Robotics . . . 10
12.3 Trend 3: Edge AI . . . 10
12.4 Trend 4: Flexible Manufacturing . . . 10
12.5 Trend 5: Lifelong Learning . . . 10

13 Market Gap **10**

14 Competitive Moat **11**

15 Market Risks **12**

16 Strategic Response to Competition **12**
16.1 1. Do Not Compete Head-On With Foundation Models . . . 12
16.2 2. Remain Robot-Agnostic . . . 12
16.3 3. Build Around Outcomes . . . 12
16.4 4. Make Integration a Product . . . 12
16.5 5. Build a Deployment Flywheel . . . 12

17 Go-to-Market Strategy **13**
17.1 Stage 1: Research Partnerships . . . 13
17.2 Stage 2: Paid Pilots . . . 13
17.3 Stage 3: Production Deployment . . . 13
17.4 Stage 4: Platform . . . 13

18 Commercial Model **14**

19 India as Initial Development Base **14**

20 Zero-to-One Evaluation **14**

21 Long-Term Market Expansion **15**

22 Final Market Verdict **15**
22.1 Strategic Position . . . 15
22.2 Final Strategic Principle . . . 15

1 Executive Summary

The robotics industry is transitioning from highly programmed automation toward increasingly adaptive and general-purpose physical intelligence.
This transition creates a potentially important software opportunity: a persistent experience layer that allows deployed robots to accumulate, organize, retrieve and apply operational knowledge over time.
QRAEMON's opportunity is therefore not primarily the creation of another robot foundation model. Its potentially differentiated position is the management of long-lived physical experience and its conversion into better future decisions.

1.1 Market Conclusion

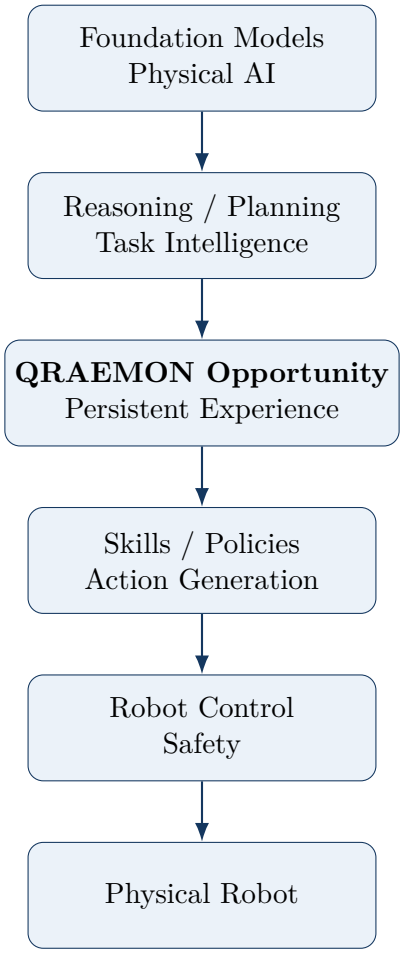
The market thesis is:

> **Robots are becoming increasingly capable, but deployed intelligence still needs to become increasingly persistent.**

The commercial opportunity exists if QRAEMON can demonstrate that accumulated experience creates measurable economic value.

2 Market Structure

The emerging robotics intelligence stack can be represented as:



QRAEMON should seek to occupy the experience layer without attempting to replace the entire stack.

3 Robotics Market Environment

3.1 Industrial Robot Base

The installed industrial robotics base is already measured in millions of operational systems globally.

This is strategically important because QRAEMON does not need to create the entire robotic hardware market.

It can potentially monetize the existing and expanding installed base.

3.2 Market Direction

Major industry trends include:

- increased automation;
- flexible manufacturing;
- AI-enabled robotics;
- vision-language-action systems;
- foundation models for physical systems;
- humanoid robotics;
- autonomous mobile robotics;
- simulation and synthetic data;
- edge AI;
- continual learning;
- robot fleet learning.

The convergence of these trends creates favorable timing for QRAEMON.

4 Physical AI Landscape

The industry is increasingly shifting from:

“AI that understands the world”

toward:

“AI that acts in the physical world.”

Important developments include foundation models and VLA systems from companies such as:

- NVIDIA;
- Google DeepMind;
- Physical Intelligence;
- Figure;

- Skild AI;
- Intrinsic;
- Covariant;
- major industrial robotics companies.

These systems create both a market tailwind and a competitive threat.

5 Competitive Landscape

5.1 Competitive Categories

Category	Representative Players	Primary Focus	Threat to QRAEMON
Robot foundation models	NVIDIA, Google DeepMind, Physical Intelligence, Skild AI	General physical intelligence	Very High
Humanoid intelligence	Figure and other humanoid companies	General-purpose humanoids	High
Robotics platforms	Intrinsic	Robot software abstraction	Very High
Warehouse robotics intelligence	Covariant and others	Task-specific robotic autonomy	High
Industrial robot OEMs	ABB, FANUC, KUKA, Universal Robots	Hardware + automation ecosystem	High
Robot memory / lifelong learning research	Academic and industrial research	Persistent adaptation	High
Traditional automation	Integrators and robot programmers	Deterministic automation	Medium

6 Key Competitor Profiles

6.1 NVIDIA

NVIDIA represents a major threat because it can combine:

- foundation models;
- simulation;
- accelerated computing;
- robotics infrastructure;
- developer ecosystem;
- deployment hardware.

The strategic implication is clear:

QRAEMON should not attempt to compete directly as another foundation-model company.

6.2 Google DeepMind

Google DeepMind's robotics work demonstrates the convergence of:

- multimodal foundation models;
- reasoning;
- robot control;
- on-device inference;
- generalization.

This creates a direct threat to broad claims of robot intelligence.
QRAEMON therefore needs a narrower and deeper technical specialization.

6.3 Physical Intelligence

Physical Intelligence is important because its thesis centers on general-purpose robot policies capable of handling many tasks and embodiments.

QRAEMON should differentiate through the temporal dimension:

What does the robot become after thousands of real-world experiences?

6.4 Figure

Figure's work demonstrates how a vertically integrated robotics company can combine:

- robot hardware;
- foundation models;
- perception;
- memory;
- manipulation;
- deployment.

The lesson is that QRAEMON's value must remain embodiment-independent.

6.5 Intrinsic

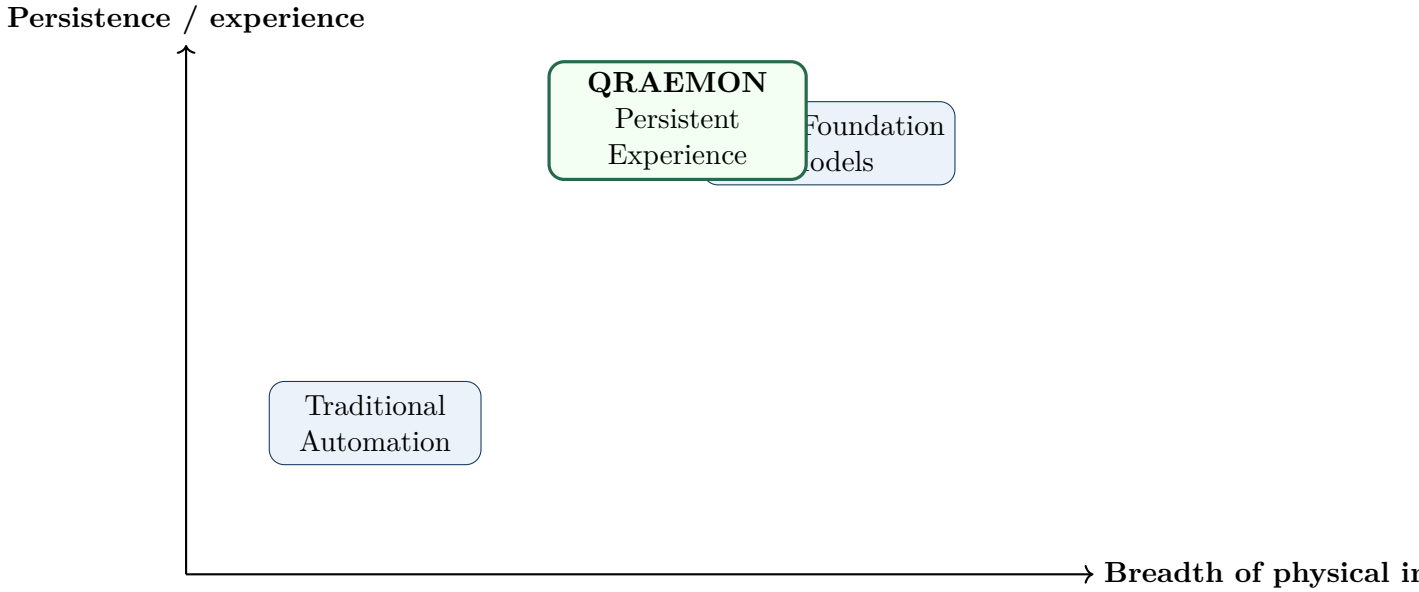
Intrinsic is particularly important because it pursues software infrastructure intended to make industrial robotics more flexible and programmable.

QRAEMON should therefore avoid becoming another generic robotics operating platform.
Its wedge should remain:

Persistent operational experience.

7 Competitive Positioning

The strategic positioning can be visualized as follows:



The objective is not necessarily to occupy the entire robotics intelligence space.
The objective is to dominate a narrow but strategically important layer.

8 Customer Segmentation

Segment	Pain	Budget	Pilot Speed	Strategic Value	Recommendation
Robotics startups	5	4	5	5	**Primary beachhead**
Industrial R&D	5	5	3	5	**Secondary beachhead**
Advanced manufacturing	5	5	2	5	Long-term
Robot integrators	4	4	4	4	Distribution partners
Research laboratories	4	2	5	4	Design partners
Humanoid companies	5	5	1	5	Strategic later market
General consumers	2	2	1	5	Avoid initially

9 Geographic Strategy

9.1 India

India is attractive for initial development because of:

- increasing industrial automation;
- expanding robotics adoption;
- lower engineering costs;
- strong technical talent;
- access to manufacturing;

- university and research ecosystems.

The initial Indian strategy should focus on:

- robotics startups;
- automotive suppliers;
- electronics manufacturing;
- advanced manufacturing;
- automation companies;
- research laboratories.

9.2 United States

The United States is strategically important for:

- robotics startups;
- AI infrastructure;
- venture capital;
- advanced manufacturing;
- humanoid robotics;
- enterprise robotics.

The long-term commercial market should therefore be global even if the initial engineering base is in India.

9.3 Europe

Europe is attractive because of:

- industrial automation;
- automotive manufacturing;
- robotics research;
- strong engineering ecosystems;
- industrial robot manufacturers.

10 Market Size Framework

A misleading approach would be:

> *"The global robotics market is $X billion, therefore QRAEMON has a $X billion market."*

QRAEMON should instead use a bottom-up model.

10.1 Installed-Robot Model

Let:

$$N = \text{number of robots addressable by QRAEMON}$$

and:

$$P = \text{annual QRAEMON revenue per robot}$$

Then:

$$TAM = N \times P$$

Illustrative scenarios:

Addressable Robots	Annual Revenue / Robot	Annual Opportunity
100,000	$5,000	$500M
250,000	$5,000	$1.25B
500,000	$5,000	$2.5B
500,000	$10,000	$5B

These are scenario calculations, not market forecasts.

They demonstrate that QRAEMON does not require ownership of the entire robotics industry to become a large software company.

11 SAM and SOM Framework

11.1 Serviceable Available Market

The initial SAM should be restricted to robots that:

- perform repeated physical tasks;
- experience meaningful environmental variation;
- generate measurable failures;
- have an accessible software interface;
- can integrate a third-party cognitive layer.

11.2 Serviceable Obtainable Market

The first SOM should be intentionally tiny.

1–3 design partners

followed by:

10–20 paid deployments

followed by:

100+ production workcells / robots

The objective is not to capture a large market immediately.

The objective is to prove a repeatable deployment mechanism.

12 Major Market Trends

12.1 Trend 1: Physical AI

AI is moving from purely digital environments into physical environments.

This creates demand for systems capable of perception, reasoning, planning and action.

12.2 Trend 2: General-Purpose Robotics

Foundation models increasingly attempt to transfer knowledge between tasks and embodiments.

This expands the potential addressable market for a robot-independent cognitive layer.

12.3 Trend 3: Edge AI

Increasing compute availability allows increasingly capable intelligence to operate locally.

This supports QRAEMON's architecture of local or embedded cognitive execution.

12.4 Trend 4: Flexible Manufacturing

Manufacturers increasingly need automation that can handle:

- product variation;
- shorter production cycles;
- higher SKU counts;
- changing environments;
- lower programming overhead.

This is particularly relevant to adaptive robotic systems.

12.5 Trend 5: Lifelong Learning

Robotics research increasingly recognizes the importance of continual adaptation, catastrophic forgetting, experience transfer and lifelong learning.

This validates the scientific direction while simultaneously increasing competition.

13 Market Gap

The potential gap can be summarized as:

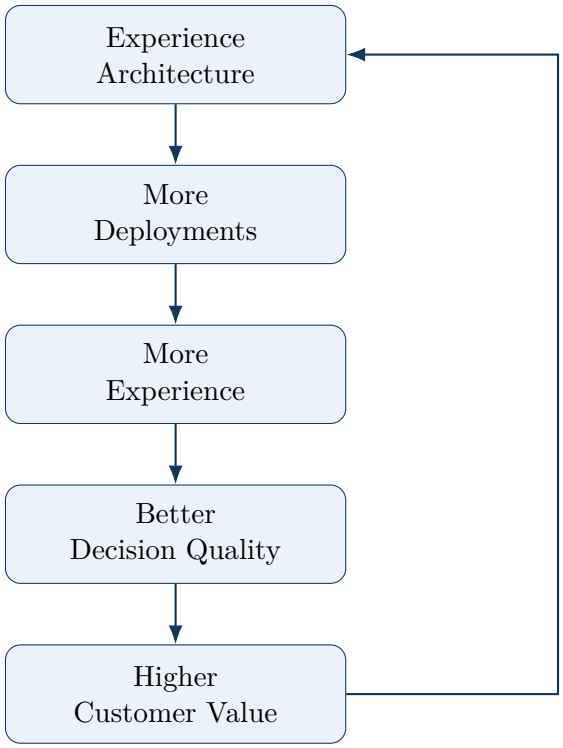


The market gap hypothesis is:

General intelligence does not automatically imply lifelong operational intelligence.
QRAEMON seeks to specialize in the latter.

14 Competitive Moat

The strongest long-term moat would be a compounding experience system.



The moat should therefore emerge from the interaction between:

1. proprietary architecture;
2. real-world experience;
3. outcome-linked memory;
4. continual adaptation;
5. cross-task transfer;
6. cross-robot transfer;
7. integration infrastructure.

15 Market Risks

Primary strategic risks

1. Foundation-model companies absorb the experience layer.
2. Robot manufacturers develop equivalent functionality internally.
3. Memory provides insufficient economic improvement.
4. Customers prefer retraining or conventional programming.
5. Integration costs become too high.
6. QRAEMON becomes a custom robotics consultancy.
7. The market remains fragmented across robot platforms.
8. Safety requirements slow deployment.
9. The architecture cannot transfer across embodiments.

16 Strategic Response to Competition

QRAEMON should follow five principles.

16.1 1. Do Not Compete Head-On With Foundation Models

Use existing foundation models where useful.

QRAEMON should own the experience architecture rather than recreate the entire intelligence stack.

16.2 2. Remain Robot-Agnostic

The core system should not depend on one robot manufacturer.

16.3 3. Build Around Outcomes

The valuable unit is not:

$$\text{stored memory}$$

but:

$$\text{stored experience} \rightarrow \text{changed decision} \rightarrow \text{improved outcome}.$$

16.4 4. Make Integration a Product

Adapters should become standardized.

16.5 5. Build a Deployment Flywheel

$$\text{Deployment} \rightarrow \text{Experience} \rightarrow \text{Better Decisions} \rightarrow \text{Better Outcomes} \rightarrow \text{More Deployments}$$

17 Go-to-Market Strategy

17.1 Stage 1: Research Partnerships

Target:

- robotics laboratories;
- robotics startups;
- university research groups.

Objective:

Technical validation

17.2 Stage 2: Paid Pilots

Target:

- robotics companies;
- industrial R&D groups;
- advanced manufacturing companies.

Objective:

Economic validation

17.3 Stage 3: Production Deployment

Convert pilots into:

- workcell licenses;
- robot licenses;
- enterprise agreements;
- fleet deployments.

17.4 Stage 4: Platform

Eventually:

QRAEMON Experience Infrastructure

becomes deployable across:

- industrial robots;
- mobile robots;
- mobile manipulators;
- humanoids;
- autonomous machines;
- scientific machines.

18 Commercial Model

Stage	Business Model	Objective
Early	Paid technical pilot	Prove value
Initial product	Per-workcell / per-robot license	Recurring revenue
Enterprise	Fleet / enterprise license	Expand account value
Platform	OEM / ecosystem licensing	Scale distribution

This is consistent with the existing QRAEMON commercial path of paid pilots followed by annual runtime licensing and eventual fleet/platform licensing. :chatgpt-content-referenceindex="1"

19 India as Initial Development Base

India should be viewed as an engineering and validation base rather than a limitation on the eventual market.

The initial strategy can exploit:

- access to engineering talent;
- robotics research institutions;
- manufacturing;
- lower experimentation costs;
- emerging industrial automation.

The eventual customer base should remain global.

20 Zero-to-One Evaluation

The QRAEMON market thesis can be evaluated using the seven-question framework in the startup evaluation worksheet.

Question	Assessment	Reasoning
Engineering 10×	3.5/5	Must prove measurable decision improvement.
Timing	5/5	Physical AI and adaptive robotics are rapidly developing.
Monopoly	3.5/5	Requires a narrow experience-management beachhead.
People	3/5	Requires AI, robotics, systems and hardware capability.
Distribution	3/5	Direct technical sales are required initially.
Durability	4.5/5	Potentially strong if experience compounds.
Secret	3.5/5	Decision-linked evolving experience is promising but unproven.

21 Long-Term Market Expansion

The intended expansion sequence should remain:

Manipulation → Mobile Robotics → Mobile Manipulation → Humanoids → Multi-Robot Systems → Other Ma

This follows the existing QRAEMON platform thesis that the cognitive architecture should persist while the physical body changes. :chatgpt-content-referenceindex="2"

22 Final Market Verdict

Market conclusion:
QRAEMON should not attempt to own the entire Physical AI market.
It should attempt to own a specific missing layer:

Persistent operational experience for physical machines.

The strongest initial commercial wedge is robotic manipulation where failures recur, human intervention is costly and existing robots already possess adequate low-level control.

22.1 Strategic Position

The strongest long-term positioning is:

QRAEMON

The Persistent Experience Layer for Physical Intelligence

The strategic progression is:

Robot Experience → Robot Recovery → Continual Robot Learning → Cross-Robot Experience → Artificial Bra

22.2 Final Strategic Principle

A robot should not have to experience the same failure twice.

This principle provides a concrete commercial entry point while remaining aligned with the larger QRAEMON objective of creating a lifelong artificial cognitive architecture.